The Estimation of a Preference-Based Measure of Health From the SF-12

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Background: The SF-12 is a multidimensional generic measure of health-related quality of life. It has become widely used in clinical trials and routine outcome assessment because of its brevity and psychometric performance, but it cannot be used in economic evaluation in its current form.

Objectives: We sought to derive a preference-based measure of health from the SF-12 for use in economic evaluation and to compare it with the original SF-36 preference-based index.

Research Design: The SF-12 was revised into a 6-dimensional health state classification (SF-6D [SF-12]) based on an item selection process designed to ensure the minimum loss of descriptive information.

Subjects: A sample of 241 states defined by the SF-6D (of 7500) have been valued by a representative sample of 611 members of the UK general population using the standard gamble (SG) technique.

Analysis: Models are estimated of the relationship between the SF-6D (SF-12) and SG values and evaluated in terms of their coefficients, overall fit, and the ability to predict SG values for all health states.

Results: The models have produced significant coefficients for levels of the SF-6D (SF-12), which are robust across model specification. The coefficients are similar to those of the SF-36 version and achieve similar levels of fit. There are concerns with some inconsistent estimates and these have been merged to produce the final recommended model. As for the SF-36 model, there is evidence of over prediction of the value of the poorest health states.

Conclusions: The SF-12 index provides a useful tool for researchers and policy makers wishing to assess the cost-effectiveness of interventions.

Key Words: SF-12, utilities, preferences, economic evaluation

(*Med Care* 2004;42: 851–859)

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Supported by GlaxoWelcome.

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ISSN: 0025-7079/04/4209-0851

Measures of health-related quality of life (HRQoL) have become widely used in clinical trials and routine outcome assessment to provide information on the effectiveness of health care. One of the most commonly used measures of HRQoL is the SF-36,[1] which has been reduced to 12 items with minimal loss of information to form the SF-12.[2] Although the psychometric properties of these instruments are well established across many conditions,[3–7] they cannot be used in their current form in economic evaluation because they do not explicitly contain preference information, nor do they generate a single index measure.

Economic evaluation in health care is increasingly concerned with comparing the cost-effectiveness of interventions in terms of their incremental cost per quality-adjusted life year,[8] and this measurement requires a single preference-based index measure of health. It would add considerable value to existing and future data sets to be able to derive such an index from the SF-12. This work presents the results of a study designed to meet this need. A valuation survey was completed to derive a utility-based algorithm for estimating a single index measure from the SF-36 for use in economic evaluation.[9] Using the same data set, this article reports on an estimation of a preference-based single index for the SF-12 and compares it with the preference-based index estimated for the SF-36.

The SF-12

The SF-12 is based on items taken from the SF-36 health survey, a standardized questionnaire used to assess patient health.[1] The SF-12 contains 12 items selected from the SF-36 on the basis of their relative efficiency or psychometric performance across 8 dimensions of health (Table 1). The SF-12 is scored to produce 2 summary scores, the physical and mental health summaries (PCS and MCS). These summary scores are estimated using a weighted formula for predicting the original SF-36 summary scores based on factor weights obtained from principle component analysis with orthogonal rotation.[2] The SF-12 summary scores have been found to be good predictors of the original SF-36 summary scores and have reproduced their psychometric performance.[10,11]

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TABLE 1. The SF-12 (Version 1.0)

Dimension	No. Items	Summary of Content	No. Response Choices	Range of Response Choice
Physical functioning	2	Moderate activities, climbing several flights of stairs	3	"Yes limited a lot" to "no, not limited at all"
Role limitations—Physical	4	Accomplishing less than wanted, limitations in the kind of activities	2	Yes/no
Bodily pain	2	Effect of pain on normal work, both inside and outside the home	6	"Not at all" to "extremely"
General health	1	Personal evaluation of health	5	"Poor" to "excellent"
Vitality	1	Feeling full of life	6	"All of the time" to "none of the time"
Social functioning	2	Extent to which physical health or emotional problems interfere with normal social activities	5	"All of the time" to "none of the time"
Role limitations—emotional	2	Accomplishing less, not working as carefully as usual	2	Yes/no
Mental health	2	Downhearted and blue, calm and peaceful	6	"All of the time" to "none of the time"

Approach

There are 3 approaches for estimating a preference-based single index measure for health from the SF-12. One approach has been to estimate values for the SF-36 by empirically mapping them onto other preference-based measures.[12,13] This indirect approach is a useful second best but is limited by the extent of the overlap of the descriptive systems of the HRQoL and preference-based measure.[14] A second approach taken by Lundberg and colleagues[15] involved administering the SF-12 alongside a self-administered preference variant of time-trade-off (TTO) in a postal survey. This study was limited by the use of a postal method for such a complex task as TTO and by the fact that the health states that were valued were not determined by statistical design but by their occurrence in a general population where the more severe states are very rare.

Another approach has been to survey the general population to value a sample of states defined by the HRQoL measure using a preference elicitation technique administered by trained interviewers. This approach has been used successfully in 2 studies to derive a preference-based index for the SF-36. One a pilot study[16] for the other study[9] that applied this approach successfully to the SF-36. This work describes a study that uses this approach to value the SF-12 and compares it with the SF-36 preference-based index.

METHODS

There are 4 components to this study. First, the SF-12 has been reduced in size and complexity so that respondents can process the information and give reliable valuations to the health states. Second, a preference-based valuation survey has been undertaken. Third, a model is estimated to predict values for all states of health described by the reduced form version of the SF-12, via alternative econometric techniques. Last, the SF-12 preference-based index is compared with the SF-36 preference-based index to see whether the former can be used in place of the latter.

The SF-6D Health State Classification

The SF-12 has 12 multilevel items many of which have no obvious ordinal relationship (Table 1); hence, many millions of health states can be defined from this classification. The valuation of such a large multiattribute function would present enormous estimation problems. It is therefore necessary to construct a simplified health state classification from the SF-12 amenable to valuation by respondents with minimal loss of descriptive information.

The number of dimensions was reduced from 8 to 6. Excluding the general health item and combining the 2 role limitation dimensions achieved this. A general health dimension is important in a profile measure such as the SF-36 because it provides a summary measure of health but is not appropriate in a health state classification designed for estimating a preference-based index. The high correlation with other dimensions means that an orthogonal design would include states where the general health statement could contradict levels on the other dimensions. A health state, for example, could describe excellent general health but have severe pain, feeling downhearted and low most of the time, being limited often in moderate activities and so forth. This presents problems for the respondent trying to imagine the state and real difficulties in estimating the impact of each

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dimension in the preference function. For this reason, none of the existing preference-based measures include general health.

It also was necessary to reduce the number of items per dimension to 1. Vitality, bodily pain, and social functioning already had 1 item each in the SF-12 (Table 1). There are 2 items for the physical functioning. The Rasch analyses undertaken by Ware and colleagues of the entire 10 items from the SF-36 PF found that the 2 SF-12 PF items were tapping the same severity of physical functioning problem.[17] Therefore, it was decided to select the more general item concerned with moderate activities in general rather than using the other item, which focused on climbing stairs.

Mental health has 2 dimensions and the decision was to use the "downhearted and low" item because this has been found to be the most important single item from the original 5-item mental health inventory for predicting mental health.[18]

The results of Rasch analyses suggested that the role items tap a similar level of severity in each role dimension and hence it was possible to select 1 from each with little loss of information. The role physical item ("were limited in the kind of work or other activities due to ...") and role emotional item ("accomplished that you would like due to...") were selected for the SF-6D because of their higher correlation with the physical and mental health summary scores generated by the SF-36 than the other role items. These 2 role items have been combined to form a 4-level SF-6D of: no limitations, limitations because of physical problems, limitations because of emotional problems, and limitations because of physical and emotional problems (Table 2).

The final version of the SF-6D uses 7 items from the SF-12 (Table 2), which is 4 less than the items used for the SF-36 version of the SF-6D—the SF-6D (SF-36). It has been named the SF-6D because it contains 6 dimensions, which is in keeping with some other preference-based measures (eg, the EQ-5D and the 15D).[27] Given its very close similarity to the SF-36 version of the SF-6D, it has been decided to retain the name and refer to the SF-12 in parenthesis. The SF-6D (SF-12) has 6 dimensions ($\delta = 1, 2,...,5$), each with between 2 and 5 levels (λ). A health state is defined by selecting 1 statement from each dimension, starting with *physical functioning* and ending with *vitality*. A total of 7500 health states

TABLE 2. The SF-6D (SF-12 version)

	Physical Functioning		**Pain**
1	Your health does not limit you in moderate activities.	1	You have pain that does not interfere with your normal work (both outside the home and housework) at all
2	Your health limits you a little in moderate activities.	2	You have pain that interferes with your normal work (both outside the home and housework) a little bit
3	Your health limits you a lot in moderate activities.	3	You have pain that interferes with your normal work (both outside the home and housework) moderately
		4	You have pain that interferes with your normal work (both outside the home and housework) quite a bit
		5	You have pain that interferes with your normal work (both outside the home and housework) extremely
	Role limitations		**Mental health**
1	You have no problems with your work or other regular daily activities as a result of your physical health or any emotional problems	1	You feel downhearted and low none of the time
2	You are limited in the kind of work or other activities as a result of your physical health	2	You feel downhearted and low a little of the time
3	You accomplish less than you would like as a result of emotional problems	3	You feel downhearted and low some of the time
4	You are limited in the kind of work or other activities as a result of your physical health and accomplish less than you would like as a result of emotional problems	4	You feel downhearted and low most of the time
		5	You feel downhearted and low all of the time
	Social functioning		**Vitality**
1	Your health limits your social activities none of the time	1	You have a lot of energy all of the time
2	Your health limits your social activities a little of the time	2	You have a lot of energy most of the time
3	Your health limits your social activities some of the time	3	You have a lot of energy some of the time
4	Your health limits your social activities most of the time	4	You have a lot of energy a little of the time
5	Your health limits your social activities all of the time	5	You have a lot of energy none of the time

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can be defined in this way. All responders to a SF-12 questionnaire can be assigned to the SF-6D (SF-12) provided the 7 items used in the 6 dimensions of the SF-6D (SF-12) have been completed.

The Valuation Survey

The valuation survey was designed to value states defined by the SF-6D (SF-36), and the data from this survey have been used here to derive a single index for the SF-6D (SF-12), which defines a subset of states from the SF-6D (SF-36). A representative sample of the general public in the United Kingdom (n = 836) valued a sample of 249 health states defined by the SF-6D (SF-36). Each respondent was asked to rank, and then value, 6 of these states using a variant of the standard gamble (SG) technique. The detailed design of the survey has been reported elsewhere.[9]

Selection of Respondents

A representative sample of the general population was achieved to reflect the variability of the population in terms of characteristics such as age, socioeconomic status, and level of education the United Kingdom. The sample was drawn using a 2-stage cluster random selection design.

Selection of Health States

The sample of states required to estimate an additive model was identified using an orthogonal design (by applying the Orthoplan procedure of SPSS), which generated 49 health states (of 18,000). It was anticipated that more complex specifications, allowing for some form of interaction between dimensions, would be estimated and therefore an additional 200 states were selected at random for inclusion in the survey. The 249 states defined on the SF-6D (SF-36) are reduced to 241 states defined by the SF-6D (SF-12) due to the simplified PF and PAIN dimensions.

Interviews

A trained and experienced interviewer conducted the interviews in the respondent's own home. The interview began with the respondent being asked to complete a short self-completion questionnaire about his or her own state of health that included completing the SF-6D (SF-36). The respondent was then asked to rank a set of 8 cards: one for each of the health states they would have to value, along with the best state defined by the SF-6D (SF-36), the worst state and immediate death.

The main part of the interview was the SG valuation of 6 health states using a variant of the SG using props developed by a team at McMaster.[18] Five SF-6D (SF-36) health states were valued against the best (full health) and the worst (PITS) health state defined by the SF-6D (SF-36). For calculating quality-adjusted life years, it is necessary to transform the results onto a scale where 1 is full health and 0 equivalent to death. Therefore, a sixth SG task was to value the PITS state against full health and death. Having valued the PITS state (P), the final step was to adjust the 5 intermediate SF-6D health state valuations (SG) onto the scale where the best SF-6D state is 1 and death 0. The health state value used in the modeling is therefore: SGADJ = SG + (1-SG) * P.

Modeling

All respondents with usable data are included whether or not they have missing values. There were 611 respondents with up to 6 valuations each (n = 3518). The models have been estimated at the mean level, that is, the explanatory variables are used to estimate the mean value given to each of the states by the respondents that valued them. Models that took account of individual variation across respondents (random effects models) gave very similar coefficient estimates to the mean models and were no better in terms of predictive ability.[9] Models are presented for the SF-12 variant of SF-6D and compared with the SF-36 variant.

A general-to-specific approach was used to derive a parsimonious regression model to estimate the mean value for each state. The explanatory variables can be classified into 2 groups. First, a set of binary dummy variables ($x_{\delta\lambda}$) that describe each level λ and dimension δ of the health state. For example, x_{31} denotes dimension $\delta = 3$ (social functioning), level $\lambda = 1$ (health limits social activities *none* of the time).

For any given health state, $x_{\delta\lambda}$ will be defined as $x_{\delta\lambda} = 1$ if, for this state, dimension δ is at level λ, and $x_{\delta\lambda} = 0$ if, for this state, dimension δ is *not* at level λ. In all cases level $\lambda = 1$ acts as the baseline for each dimension.

Second, there is a binary dummy variable to take account of any additional effect on health state value when 1 or more dimension of health is at the "most severe" level. "Most severe" is defined as level 3 for physical functioning; levels 3 and 4 for role limitation; and levels 4 and 5 for social functioning, pain, and mental health; and level 5 for vitality.

In both cases the intercept in the regression model is restricted to equal unity. In theory, this term represents the value of full health, ie, when each dimension of the heath state is at level 1. However, in practice estimates of the intercept are usually less than 1.[9,19] The techniques used in the valuation survey are based on the assumption that state 111111 is to equal 1 and death is equal to 0. For state 111111 to hold any other value would change the scale, hence the restriction is imposed.

The models presented in this article were estimated by ordinary least square regressions with the MOST term included to account for interactions. Explanatory power is expressed in terms of an adjusted R^2. However, the overall aim is to predict health state values and this has been assessed in terms of mean absolute error and the proportion of predictions outside 0.05 and 0.1 ranges on either side of the actual value. Predictions are further tested in terms of bias (t test)

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and whether or not the prediction errors are normally distributed (JB test). Finally, the presence of autocorrelation in prediction errors is tested using the Ljung-Box statistic by ordering errors in terms of actual health state values. All analysis was conducted in Stata 7 (StataCorp LP, College Station, TX) and Eviews 4 (QMS, Irvine, CA).

Responsiveness Compared With the SF-36 Preference-Based Index

The responsiveness of the SF-12 preference-based index is compared with the SF-36 index across 7 patient groups obtained from 4 studies.[5,20–22] These studies all used the SF-36 to assess health status on at least 2 occasions. The patient groups were chronic obstructive pulmonary disease, venous leg ulcers (with and without a healed ulcer), osteoarthritis (total knee replacement), osteoarthritis (medical), and a sample of elderly people participating in an exercise trial (intervention and control arms). Longitudinal health status changes were assessed differently for each group. Chronic obstructive pulmonary disease and osteoarthritis medical patients were characterized in terms of their response to the SF-36 health transition item at 6-month follow-up.[5,20] Leg ulcer patients have been divided into those who responded to treatment and those who did not.[22] The osteoarthritis surgical group were assessed before and after a total knee replacement operation.[20] The elderly exercise group were divided into those in the exercise arm of the trial and those in the control and followed up for 12 months.[22]

Longitudinal change has been assessed in terms of the mean change and the standardized response mean (SRM) of the change.[23] The SRM is a version of the effect size that divides the mean change by the standard deviation of the change.

RESULTS

The Sample

Of the 1445 addresses contacted for interview, 167 proved to be ineligible and 836 were successfully interviewed (a 65% response rate). One hundred and thirty respondents had to be excluded from the analysis for failing to value the PITS state; therefore, it was not possible to generate an adjusted SG value (see below). A further 9 were excluded for not valuing 2 or more health states. Finally, there were 86 respondents whose health state values did not change between the 5 states. This last group has been excluded because the lack of variation is likely to indicate that the respondent did not understand the task.

Table 3 shows that the final sample of 611 respondents is broadly representative of the UK general population of adults. The mean age is the same, but our sample had a slightly larger proportion who are aged older than 65, more likely to be female, slightly less likely to be in employment, and more likely to have no educational qualifications. The health status of the sample using the SF-6D preference-based index was just 0.03 lower than UK general population.

TABLE 3. Characteristics of Included Respondents and UK Population

	Included Respondents (n = 611)	UK Population
Age (mean adult)	46	46
% aged 65 and older	19	15
% female	59	51
% married	53	50
% with children < 16	28	29
% employed	55	60
% with university degree	16	16
% with no educational qualifications	30	23
Mean (SD) SF-6D score	0.767 (0.113)	0.799

UK population sources are the General Household Survey (www.statistics.gov.uk) and the British Household Panel Survey 1998 (ESRC Data Archive, www.data-archive.ac.uk).

Of the 611 individuals included in the data set, there were 148 missing values from 117 individuals. This results in 3518 observed SG valuations across 241 uniquely defined health states and these form the data set reported and analyzed below.

The Data Set

Each of the 241 health states has been valued an average of 15 times. Mean health state values range from 0.10 to 0.99 and generally have large standard deviations. The relative health state valuations broadly conform to the logical ordering of the SF-6D. Most respondents valued the worst health state as better than death (445/611). However, very few health states were valued at 1.0 (20/3518), indicating the willingness of respondents to risk a worse health state to have the chance of a better state of health.

Four models are reported in Table 4. Models (1) and (3) are the preferred full models for both the SF-6D (SF-36) and SF-6D (SF-12), whereas models (2) and (4) are the parsimonious consistent models. The negative sign on the interaction term (MOST) reflects the additional loss of utility when a state contains at least one dimension at the most severe level.

The consistent models were constructed by removing variables that were not significant at $P < 0.05$. In addition levels of each dimension were aggregated if inconsistencies occurred; that is, if the coefficients on each level did not represent increasing decrements to health state value. For example, comparing models (3) and (4) the insignificant dimensions at the top end of the PF and PAIN dimensions have been removed. Also the inconsistent estimates in the

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TABLE 4. Comparing the SF-6D and SF-12: Full and Consistent Models at Mean Level

	SF-6D(SF-36)			SF-6D(SF-12)		
	(1)		(2)	(3)		(4)
c	**1.000**	c	**1.000**	c	**1.000**	c	**1.000**
PF2	−0.053						
PF3	−0.011	**PF23**	**−0.035**	PF2	0.012		
PF4	**−0.040**	**PF4**	**−0.044**				
PF5	**−0.054**	**PF5**	**−0.056**	PF3	**−0.040**	**PF3**	**−0.045**
PF6	**−0.111**	**PF6**	**−0.117**				
RL2	**−0.053**			RL2	**−0.068**		
RL3	**−0.055**	**RL234**	**−0.053**	RL3	**−0.061**		
RL4	**−0.050**			RL4	**−0.054**	**RL234**	**−0.063**
SF2	**−0.055**	**SF2**	**−0.057**	SF2	**−0.059**	**SF2**	**−0.063**
SF3	**−0.067**	**SF3**	**−0.059**	SF3	**−0.069**	**SF3**	**−0.066**
SF4	**−0.070**	**SF4**	**−0.072**	SF4	**−0.078**	**SF4**	**−0.081**
SF5	**−0.087**	**SF5**	**−0.087**	SF5	**−0.093**	**SF5**	**−0.093**
PAIN2	**−0.047**						
PAIN3	−0.025	**PAIN23**	**−0.042**	PAIN2	−0.004		
PAIN4	**−0.056**	**PAIN4**	**−0.065**	PAIN3	−0.039	**PAIN3**	**−0.042**
PAIN5	**−0.091**	**PAIN5**	**−0.102**	PAIN4	**−0.076**	**PAIN4**	**−0.077**
PAIN6	**−0.167**	**PAIN6**	**−0.171**	PAIN5	**−0.140**	**PAIN5**	**−0.137**
MH2	**−0.049**			MH2	**−0.063**		
MH3	**−0.042**	**MH23**	**−0.042**	MH3	**−0.055**	**MH23**	**−0.059**
MH4	**−0.109**	**MH4**	**−0.100**	MH4	**−0.119**	**MH4**	**−0.113**
MH5	**−0.128**	**MH5**	**−0.118**	MH5	**−0.140**	**MH5**	**−0.134**
VIT2	**−0.086**			VIT2	**−0.097**		
VIT3	**−0.061**			VIT3	**−0.065**		
VIT4	**−0.054**	**VIT234**	**−0.071**	VIT4	**−0.059**	**VIT234**	**−0.078**
VIT5	**−0.091**	**VIT5**	**−0.092**	VIT5	**−0.103**	**VIT5**	**−0.106**
MOST	**−0.070**	**MOST**	**−0.061**	MOST	**−0.085**	**MOST**	**−0.077**
n	249		249		241		241
adj R^2	0.526		0.472		0.427		0.433
Predictive Ability							
MAE	0.073		0.074		0.079		0.079
No. > |0.05|	120		116		121		125
No. > |0.10|	51		58		58		55
t	−1.146		−1.317		−1.647		−1.780
JB	0.173		1.169		0.964		1.650
LB	**189.87**		**163.87**		**122.49**		**111.57**

Models are estimated with White heteroscedasticity consistent standard errors.
Figures in bold are significant at $t_{0.10}$.
MAE indicates mean absolute error.
t is a test that the mean of the prediction errors is zero; JB, Jarque-Bera test for normality of errors; LB, Ljung-Box test for autocorrelation of errors.

ROLE, VIT, and MH dimensions have been aggregated to achieve consistent scales. The parsimonious consistent models are the preferred specification and these results are discussed further below.

The explanatory power of models (2) and (4) is comparable. For all dimensions except physical functioning and pain, the SF-6D (SF-12) model attracts larger coefficients than the SF-6D (SF-36). In both models PAIN appears to be the most important dimension in determining health state values, followed by mental health and, in the case of the SF-6D (SF-36), physical functioning. However, the limited physical functioning dimension of the SF-12 reduces its importance.

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These results can be used to estimate values for each state. For example, the estimated value for state 321335 is 0.675 from the SF-6D (SF-36) model (1 − 0.035 − 0.053 − 0.042 − 0.042 − 0.092 − 0.061) and 0.695 from the SF-6D (SF-12) model. The statistics at the bottom of Table 4 judge the overall within sample predictive ability of the models. The mean absolute errors across all predictions are small; in addition, the prediction errors are unbiased and normally distributed. For both models, approximately one fifth of predictions are within ±0.05 of the actual value, and approximately one half are within ±0.10. However, there is a systematic pattern in the prediction errors from both models (Ljung-Box statistic), which reflects a tendency to overpredict the value of poor health states. This is demonstrated by the line plot in Figure 1, which shows actual and predicted values for both models ordered by mean actual values.

Table 5 presents the results of comparing the responsiveness of the new SF-12 and the SF-36 preference-based indices. Across the 7 patient groups obtained from the 4 studies, it can be seen that mean difference estimated by the SF-12 index and its SRM are never smaller than those produced by the SF-36 index. Indeed in some cases they are slightly larger.

DISCUSSION AND CONCLUSION

The results of this study provide a way of estimating a preference-based single index from existing and future SF-12 databases. It also offers an alternative to existing preference-based measures, such as the EQ-5D[24] or the HUI3,[28] where a researcher has good psychometric grounds for believing the SF-12 provides a better description of the impact of the health care intervention.

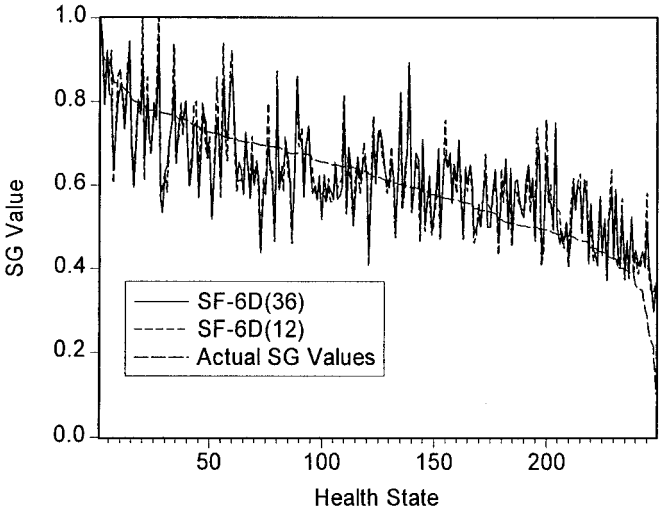

FIGURE 1. Plot of actual and predicted values for models (2) and (4).

The SF-12 index requires 7 of the 12 items of the instrument to be completed, which are fewer than the SF-36 preference-based index. However, evidence from applying the SF-12 index algorithm to 7 patient groups experiencing modest health changes suggests that it is just as responsive to change and generates similar estimates to the SF-36 preference-based index.

The SF-12 preference-based index is unlikely to entirely replace existing preference-based measures. It has been known that the SF-36, and its predecessor the SF-20, suffer from a floor effect,[27] and this has been confirmed in more recent comparisons of the SF-6D with the EQ-5D and HUI3.[28–30] This has been made slightly worse for the SF-12 index because the model overpredicts the poorer states.

The existence of a floor effect is reflected in the proportion that regarded the worse state as better than death, which for the SF-6D was 73% compared with 9% for HUI3 and 8% for EQ-5D. The floor effect is also reflected in the range of index values generated by each preference-based measure, with the lowest value for the SF-6D being 0.35 compared with −0.36 for the HUI3[28] and −0.59 for the Measurement and Valuation of Health (MVH) TTO algorithm for the EQ-5D.[20] For severe conditions, the EQ-5D and the HUI3 are likely to be more appropriate measures for generating preference-based index numbers for use in economic evaluation. By contrast, the EQ-5D would appear to suffer from a ceiling effect, with large numbers of respondents reporting no problems, hence for milder conditions the SF-6D may have an important role to play[28,30]

The main weaknesses from this study arise from the attempt to value a comparatively large classification describing 7500 health states. Despite this, the main effect estimates were found to be robust across model specifications and similar to those for the SF-36 version of SF-6D. In most cases they were also consistent with ordinal levels of the SF-6D. Of course, a larger sample size and the valuation of additional health states may have overcome some of the inconsistencies and nonsignificant coefficients for some of the dimension levels. The explanatory power of the preferred mean model[4] presented in this paper of 0.43 is similar to the York MVH TTO model for the EQ-5D of 0.45,[20] although the mean absolute error was larger at 0.079 compared with 0.039. A Comparison between these 2 pieces of work is difficult because the valuation of the SF-6D is much larger undertaking describing nearly 75 times more states. By comparison the HUI3 achieved a mean absolute error of 0.067 within sample and 0.087 out of sample.[26]

There might be a concern that the valuation survey has been restricted to the United Kingdom and so it is important to extend the valuation work to other countries, including the United States. Another concern is with the existence of systematic prediction errors in the model. We have attempted numerous other specifications for interactions not reported in

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TABLE 5. Responsiveness of New SF-12 Index Compared With SF-36 Index

Patient Group	n	Longitudinal Change	Mean Score Change in Index		Standardized Response Mean	
			SF12	SF36	SF12	SF36
Chronic obstructive pulmonary disease[5]	44	Health transition question	0.0136 (0.076)	0.0136 (0.099)	0.18	0.13
Leg ulcer						
Healed	85	Change over 12 months	−0.028 (0.155)	−0.013 (0.148)	0.18	0.09
Unhealed[22]	69		−0.059 (0.151)	−0.052 (0.142)	0.39	0.37
Osteoarthritis (total knee replacement group)[20]	87	Knee replacement	0.045 (0.119)	0.033 (0.102)	0.38	0.32
Osteoarthritis (medical group)[20]	94	Health transition question	0.040 (0.103)	0.040 (0.104)	0.39	0.38
Elderly exercise trial						
No exercise group	3693	Change over 12 months	−0.014 (0.113)	−0.012 (0.100)	0.12	0.12
Exercise group[21]	1873		−0.007 (0.114)	−0.004 (0.100)	0.06	0.04

this article, but these were not able to improve on the results reported here.

This study is the first attempt to estimate a preference-based index using general population sample to value most of the descriptive system of the SF-12. It confirms findings from psychometrics analysis that using the SF-12 results in little loss of information compared with the SF-36. These results can be applied to any SF-12 data to generate a preference-based single index for use in economic evaluation and other studies requiring a single index.

ACKNOWLEDGMENTS

We would like to thank Roger Thomas and Patrick Sturgis at SCPR for conducting the valuation survey and GlaxoSmithKline for funding the study. We would also like to thank John Ware and Barbara Gandek from the Health Assessment Laboratory in Boston for their help and support throughout this study and to Mark Deverill for his contribution to the project. The usual disclaimer applies.

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